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WATER BLOCK PRESSURE REGULATION STRUCTURE

Abstract

A water block pressure regulation structure includes a top cover internally defining a water-out compartment, a pressure reduction chamber and a water-in compartment; a heat absorbing member; and a partitioning plate superposed on the heat absorbing member to define a heat exchange chamber between them, and including a buffer space communicable with the heat exchange chamber via first and second bores formed on the partitioning plate. The first and second bores are openable and closeable by first and second valves fitted therein, respectively. The water-in and water-out compartments communicate with the heat exchange chamber via first and second penetrating openings on the partitioning plate. When the heat exchange chamber internally has high pressure or insufficient cooling liquid, the first and second valves automatically open to communicate the buffer space with the heat exchange chamber, and accordingly regulate the pressure and cooling liquid in the heat exchange chamber.

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Background/Summary

[0001] This application claims the priority benefit of Taiwan patent application number 113108707 filed on Mar. 8, 2024.

FIELD OF THE INVENTION

[0002] The present invention relates to the field of water block, and more particularly, to a water block pressure regulation structure.

BACKGROUND OF THE INVENTION

[0003] The existing water cooling system includes some prerequisite components, such as water block, water radiator, water pump and pipes. The water block, the water radiator and the water pump are connected to one another by the pipes to form a water liquid circulation path in the form of a closed circuit. The water block is in contact with a heat source to absorb heat produced by the heat source. The absorbed heat is finally carried away by a working fluid to achieve the effect of heat exchange. The water pump is mainly used to drive the working fluid to circulate through the whole closed circuit; and the water radiator is a unit mainly used to enhance the cooling efficiency of the working fluid in the whole circulation path.

[0004] When the working fluid (i.e. a cooling liquid), such as water, circulates to achieve the purpose of cooling, the cooling liquid would have an increased volume due to thermal expansion under increasing temperature of the cooling liquid and accordingly, reduced density. When the cooling liquid filled in the closed circuit expands in volume, it would possibly affect different components in the water cooling system to result in burst and leakage of the water cooling system. For example, the water block or the water radiator might be pushed by the expanded cooling liquid to result in deformed chamber thereof, or the joints of different components with the pipes might burst to separate from each other.

[0005] Following problems will occur when the above described states happened: [0006] (1) The water block is deformed and fails to tightly contact with the heat source to thereby result in thermal resistance; [0007] (2) The joints of different components with the pipes in the water cooling system burst to result in separation of components from the pipes and accordingly, leaking of cooling liquid; and [0008] (3) The water radiator would deform to result in poor heat dissipation and accordingly, poor cooling efficiency of the cooling liquid.

[0009] Specifically, any one of the above conditions would lead to failure of the whole water cooling system, which would in turn cause damages or even burn-out of the electronic device or electronic equipment that uses the water cooling system to dissipate heat produced during operation thereof.

[0010] Presently, to overcome the above problems, there are manufacturers who provide a pressure relief valve in the water cooling system set to open when the cooling liquid produces increased pressure in a temperature rise condition, so as to balance the internal pressure of the water cooling system automatically. However, the above solution causes another problem, that is, the cooling liquid in the water cooling system would decrease but could not be replenished automatically. When the cooling liquid decreases gradually, the cooling efficiency of the whole water cooling system will drop; and once the cooling liquid is used up, the whole water cooling system would stop working.

[0011] It is therefore tried by the inventor to overcome the problems and disadvantages in the prior art water cooling system by developing a water block pressure regulation structure.

SUMMARY OF THE INVENTION

[0012] To overcome the problems in the prior art water cooling system, a primary object of the present invention is to provide a water block pressure regulation structure capable of automatically regulating the pressure in the water cooling system.

[0013] Another object of the present invention is to provide a water block pressure regulation structure that enables a cooling liquid to flow back to the water cooling system automatically.

[0014] To achieve the above and other objects, the water block pressure regulation structure of the present invention includes an upper cover, a partitioning plate, and a heat absorbing member.

[0015] The top cover has a pressure reduction chamber and an outlet and an inlet; the pressure reduction chamber includes a water-out compartment and a water-in compartment located at positions corresponding to the outlet and the inlet, respectively; and the partitioning plate is superposed on the heat absorbing member to form a heat exchange chamber between them.

[0016] The partitioning plate is provided on one side facing toward the top cover with an upward extended valve receiving section consisting of communicable first zone and second zones. The pressure reduction chamber in the top cover correspondingly encloses the first and the second zone to form an independent buffer space around an outer side of the first and second zones. The partitioning plate is provided with a first and a second bore in the first and the second zone, respectively; and the heat exchange chamber and the buffer space are communicable with each other via the first and the second bore. The partitioning plate further has a first valve and a second valve provided thereon to be set in the first and the second zone, respectively, for opening or closing the first and the second bore, so as to control the communication between the heat exchange chamber and the buffer space. The partitioning plate is provided near the valve receiving section with a first and a second penetrating opening, via which the inlet and outlet of the top cover are communicable with the heat exchange chamber to form a cooling liquid circulation path.

[0017] When the cooling liquid in the heat exchange chamber is heated and expanded to excessively increase the pressure in the heat exchange chamber, and when the cooling liquid in the heat exchange chamber is insufficient, the first and the second hole can be opened by the first and the second valve, respectively, so as to communicate the buffer space with the heat exchange chamber to enable regulation of pressure and replenishment of cooling liquid in the heat exchange chamber.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The structure and the technical means adopted by the present invention to achieve the above and other objects can be best understood by referring to the following detailed description of the preferred embodiment and the accompanying drawings, wherein

[0019] FIG. 1A is a top exploded perspective view of a water block pressure regulation structure of the present invention;

[0020] FIG. 1B is a bottom exploded perspective view of the water block pressure regulation structure of the present invention;

[0021] FIG. 1C is an assembled perspective view of the water block pressure regulation structure of the present invention;

[0022] FIG. 2A is a cutaway view taken along line a-a of FIG. 1C;

[0023] FIG. 2B is a cutaway view taken along line b-b of FIG. 1C;

[0024] FIG. 3A is a sectional view showing how a cooling liquid flows in the water block of the present invention;

[0025] FIG. 3B is a sectional view showing the water block pressure regulation structure of the present invention in working; and

[0026] FIG. 3C is another sectional view showing the water block pressure regulation structure of

the present invention in working.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0027] The present invention will now be described with a preferred embodiment thereof.

[0028] Please refer to FIGS. 1A, 1B and 1C, which are top exploded perspective view, bottom exploded perspective view and assembled perspective view, respectively, of a water block pressure regulation structure according to an embodiment of the present invention; and to FIGS. 2A and 2B, which are cutaway views taken along line a-a and line b-b of FIG. 1C, respectively. As shown, the present invention includes a water block structure **1**, which has a top cover **11**, a partitioning plate **12**, and a heat absorbing member **13**.

[0029] The top cover **11** is a hollow member consisting of an outer shell **111**, a pressure reduction chamber **112**, and an outlet **113** and an inlet **114**. The outer shell **111** is located at an outer side of the top cover **11**, and the pressure reduction chamber **112** is located at an inner side of the top cover **11**. A water-out compartment **115** and a water-in compartment **116** are formed in the pressure reduction chamber **112** corresponding to and communicable with the outlet **113** and the inlet **114**, respectively. The top cover **11** is provided at an open end thereof with a stepped section **117**, as indicated in FIG. 3A, which will be discussed in detail later.

[0030] The partitioning plate **12** has a first side **121** facing toward the top cover **11** and an opposite second side **122** facing toward the heat absorbing member **13**. The partitioning plate **12** is provided on one side (it is the first side **121** in the illustrated embodiment) with an upward extended valve receiving section **123** enclosing a first zone **1231** and a second zone **1232**, which are communicable with each other. The partitioning plate **12** is further provided with a first bore **1233** and a second bore **1234** in the first and the second zone **1231**, **1232**, respectively, to penetrate the first side **121** and the second side **122** of the partitioning plate **12**. An area on the partitioning plate **12** near outer edges of the first side **121** is in contact at with the stepped section **117** of the top cover **11**.

[0031] The first zone **1231** is internally provided with a first valve **14** and an elastic element **15**. The first valve **14** has two ends respectively being a protruded portion **141** facing toward the partitioning plate **12** for correspondingly fitting in the first bore **1233**, and a recessed portion **142** facing toward the top cover **11** for pressing against the elastic element **15**, such that the first valve **14** can open or close the first bore **1233**. The second zone **1232** is internally provided with a second valve **16**, which includes a valve main body **16**, a limiting protrusion **162**, and a stopper portion **163**. The limiting protrusion **162** and the stopper portion **163** are located at two different sides of the partitioning plate **12**. The valve main body **161** upwardly extends through the second bore **1234**, such that the stopper portion **163** covers the second bore **1234**, which is openable or closeable via the second valve **16**. The partitioning plate **12** is provided on the second side **122** with a recess **124** for receiving the stopper portion **163** therein. The partitioning plate **12** is also provided near two opposite sides of the valve receiving section **123** with a first penetrating opening **125** and a second penetrating opening **126**, both of the first and the second penetrating opening **125**, **126** extend through the partitioning plate **12**.

[0032] The heat absorbing member **13** has a heat dissipation surface **131** and a heat absorption surface **132** located at an upper side and a lower side of the heat absorbing member **13**, respectively. The heat dissipation surface **131** has a plurality of radiating fins **1311** provided thereon, and has a heat exchange flow passage **1312** formed between two adjacent radiating fins **1311**. The partitioning plate **12** is superposed on the heat absorption member **13**, and both the partitioning plate **12** and the heat absorption member **13** are covered in the outer shell **111** of the top cover **11**, such that the top cover **11**, the partitioning plate **12** and the heat absorption member **13** are joined with each other to form a water block structure **1**.

[0033] The pressure reduction chamber **112** in the top cover **11** is located on an outer side of the first and the second zone **1231**, **1232** to cover them correspondingly, such that an independent buffer space **1235** is defined between the top cover **11** and the valve receiving section **123**, and a heat exchange chamber **133** is defined between the heat dissipation surface **131** of the heat

absorption member **13** and the second side **122** of the partitioning plate **12**. The buffer space **1235** and the heat exchange chamber **133** are communicable with each other via the first and the second bore **1233**, **1234**, while the first and the second bore **1233**, **1234** are controlled by the first and the second valve **14**, **16** to open or close.

[0034] The water-in and the water-out compartments **116**, **115** are communicable with the heat exchange chamber **133** via the first and the second penetrating opening **125**, **126**, respectively, such that the heat exchange chamber **133**, the water-in compartment **116**, and the water-out compartment **115** together form a complete cooling liquid circulation path. When a cooling liquid in the heat exchange chamber **133** is heated and expanded, the first valve **14** is pushed by increased pressure to open the first bore **1233**, allowing a part of the cooling liquid to flow from the heat exchange chamber **133** through the first bore **1233** into the buffer space **1235**. When the temperature of the cooling liquid drops and the pressure generated by the cooling liquid is reduced, the second valve **16**, which initially closes the second bore **1234** when being pushed by pressure, is now released from the pressure and influenced by a gravitational force of the cooling liquid to open the second bore **1234**; and the cooling liquid previously entered the buffer space **1235** flows through the second bore **1234** back to the heat exchange chamber **133** again.

[0035] To ensure good water tightness of the water block structure **1** and prevent the cooling liquid from leakage, a waterproof seal gasket **17** is provided at the stepped section **117**, lest cooling liquid in the water block structure **1** should leak from the joint between the partitioning plate **12** and the heat absorbing member **13**.

[0036] The water block structure **1** of the present invention may further include a fixing frame **2**, which securely frames the top cover **11**, the heat absorbing member **13**, and the partitioning plate **12** simultaneously while the fixing frame **2**, the top cover **11**, the heat absorbing member **13**, and the partitioning plate **12** are locked together along their outer edges by a plurality of fastening elements, such as screws, so that the water block structure **1** can be tightly locked and flatly attached to at least one heat source (not shown).

[0037] Please refer to FIG. 3A along with other figures. The arrows in FIG. 3A indicate a flowing direction of cooling liquid **4**. As shown, the cooling liquid **4** flows from a pipe **3** into the communicable inlet **114** and water-in compartment **116** sequentially, and then flows into the heat exchange chamber **133**. The cooling liquid **4** passing through the heat exchange chamber **133** rises in temperature gradually, and keeps flowing through the water-out compartment **115** to the outlet **113** and leave the water block structure **1** via another pipe **3**.

[0038] Please refer to FIG. 3B, which shows the state of the water block structure **1** after the cooling liquid **4** is heated. When the cooling liquid **4** is heated and expanded (to have a reduced density), it generates outward expanded pressure against the heat exchange chamber **133**. At this point, the pressure pushes the first and the second valve **14**, **16** upward, such that the first valve **14** subjected to the pressure pushed upward against the elastic element **15**, allowing the cooling liquid **4** to flow from the heat exchange chamber **133** through the first bore **1233** into the buffer space **1235**, and the second valve **16** subjected to the pressure is moved upward by the pressure. At this point, the stopper portion **163** of the second valve **16** is fully attached to the recess **124** on the second side of the partitioning plate **12** to close the second bore **1234**.

[0039] Please refer to FIG. 3C, which shows the state of the water block structure **1** after the cooling liquid is cooled. As shown, when the cooling liquid **4** is lowered in temperature, it returns to its normal volume and generates less pressure. At this point, the elastic element **15** pushes the first valve **14** toward the partitioning plate **12** to close the first bore **1233**; and the second valve **16** naturally moves down toward the heat exchange chamber **133** when the cooling liquid **4** flows downward under the gravitational force, such that the limiting protrusion **162** on the second valve **16** is in contact with the first side **121** of the partitioning plate **12**, and a gap is formed between the stopper portion **163** and the recess **124**, allowing the cooling liquid **4** in the buffer space **1235** to flow through the gap in the second bore **1234** back to the heat exchange chamber **133**.

[0040] In other words, according to the present invention, the first valve **14** and the second valve **16** is changeable in their positions (i.e. can be pushed upward or fall downward) with the temperature rise and drop of the cooling liquid **4**. In the present invention, the heat exchange chamber **133** and the buffer space **1235** are communicable via the first bore **1233** and the second bore **1234**. When the cooling liquid **4** is heated and expanded to have an increased volume, the additional volume of the cooling liquid **4** will flow into the buffer space **1235**. On the other hand, when the cooling liquid **4** is lowered in temperature, the cooling liquid **4** previously stayed in the buffer space **1235** will flow back to the heat exchange chamber **133** again.

[0041] In the above process, no additional operation is needed to handle the water block structure **1**. The first and second valves **14**, **16** of the water block structure **1** will move automatically to regulate the pressure in the water block structure **1** along with the rise or drop of the temperature of the heat source, so that the water block structure **1** is protected against unexpected damage due to overpressure in the cooling system.

[0042] In conclusion, the present invention has the following advantages: [0043] (1) Being capable of regulating the internal pressure of the water block structure automatically; [0044] (2) Without the need of replenishing the cooling liquid; and [0045] (3) Having simple and reliable structure.

[0046] The present invention has been described with a preferred embodiment thereof and it is understood that many changes and modifications in the described embodiment can be carried out without departing from the scope and the spirit of the invention that is intended to be limited only by the appended claims.

Claims

1. A water block pressure regulation structure, comprising: a top cover having a pressure reduction chamber and an outlet and an inlet; the pressure reduction chamber including a water-out compartment and a water-in compartment located at positions corresponding to the outlet and the inlet, respectively; a partitioning plate being provided on one side with an upward extended valve receiving section consisting of a first zone and a second zone, which are communicable with each other; the partitioning plate being further provided with a first bore and a second bore in the first and the second zone, respectively; the first zone having a first valve and an elastic element provided therein for opening or closing the first bore and the second zone having a second valve provided therein for opening or closing the second bore; the partitioning plate being further provided outside of the valve receiving section with a first penetrating opening and a second penetrating opening; and the pressure reduction chamber in the top cover correspondingly covering the first and the second zone to form an independent buffer space around an outer side of the first and the second zone; and a heat absorbing member having a heat absorption surface and a heat dissipation surface; the partitioning plate being superposed on the heat absorbing member with a heat exchange chamber defined between the heat dissipation surface of the heat absorbing member and the partitioning plate; the heat exchange chamber being communicable with the water-in compartment and the water-out compartment via the inlet and the outlet, respectively, to form a cooling liquid circulation path; the buffer chamber and the heat exchange chamber being independent of each other but communicable via the first and the second bore only; wherein when the heat exchange chamber has a relatively high internal pressure or when a cooling liquid in the cooling liquid circulation path is insufficient, the first and the second valve automatically open or close the first and the second bore, respectively, to achieve regulation of pressure and replenishment of the cooling liquid in the heat exchange chamber.

2. The water block pressure regulation structure as claimed in claim 1, wherein the partitioning plate has a first side facing toward the top cover and a second side facing toward the heat absorbing member, and the heat exchange chamber being defined between the second side and the heat dissipation surface; and the first and the second penetrating opening extending through the

partitioning plate.

3. The water block pressure regulation structure as claimed in claim 2, wherein the first valve being set in the first zone to close the first hole, and the elastic element being disposed in the first zone with two ends elastically pressed against the first valve and the top cover.
 4. The water block pressure regulation structure as claimed in claim 2, wherein the first valve includes a protruded portion and a recessed portion; the protruded portion being faced toward the partitioning plate for correspondingly fitting in the first bore; and the recessed portion being faced toward the top cover with one end of the elastic element pressed against the recessed portion.
 5. The water block pressure regulation structure as claimed in claim 2, wherein the second valve includes a valve main body, and a limiting protrusion and a stopper portion to be located at two different sides of the partitioning plate; and the valve main body extending through the second bore, such that the stopper portion covers the second bore.
 6. The water block pressure regulation structure as claimed in claim 5, wherein the partitioning plate is provided on the second side with a recess for receiving the stopper portion therein.
 7. The water block pressure regulation structure as claimed in claim 2, wherein the top cover has a stepped section formed at a lower open end thereof; and an area on the partitioning plate near outer edges of the first side being in contact with the stopped section.
 8. The water block pressure regulation structure as claimed in claim 1, wherein the heat absorbing member is provided on the heat dissipation surface with a plurality of radiating fins, and a heat exchange flow passage being formed between any two adjacent radiating fins.
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